# Machine Learning:

Cost function , Greadent Decent Global and Local Minima .

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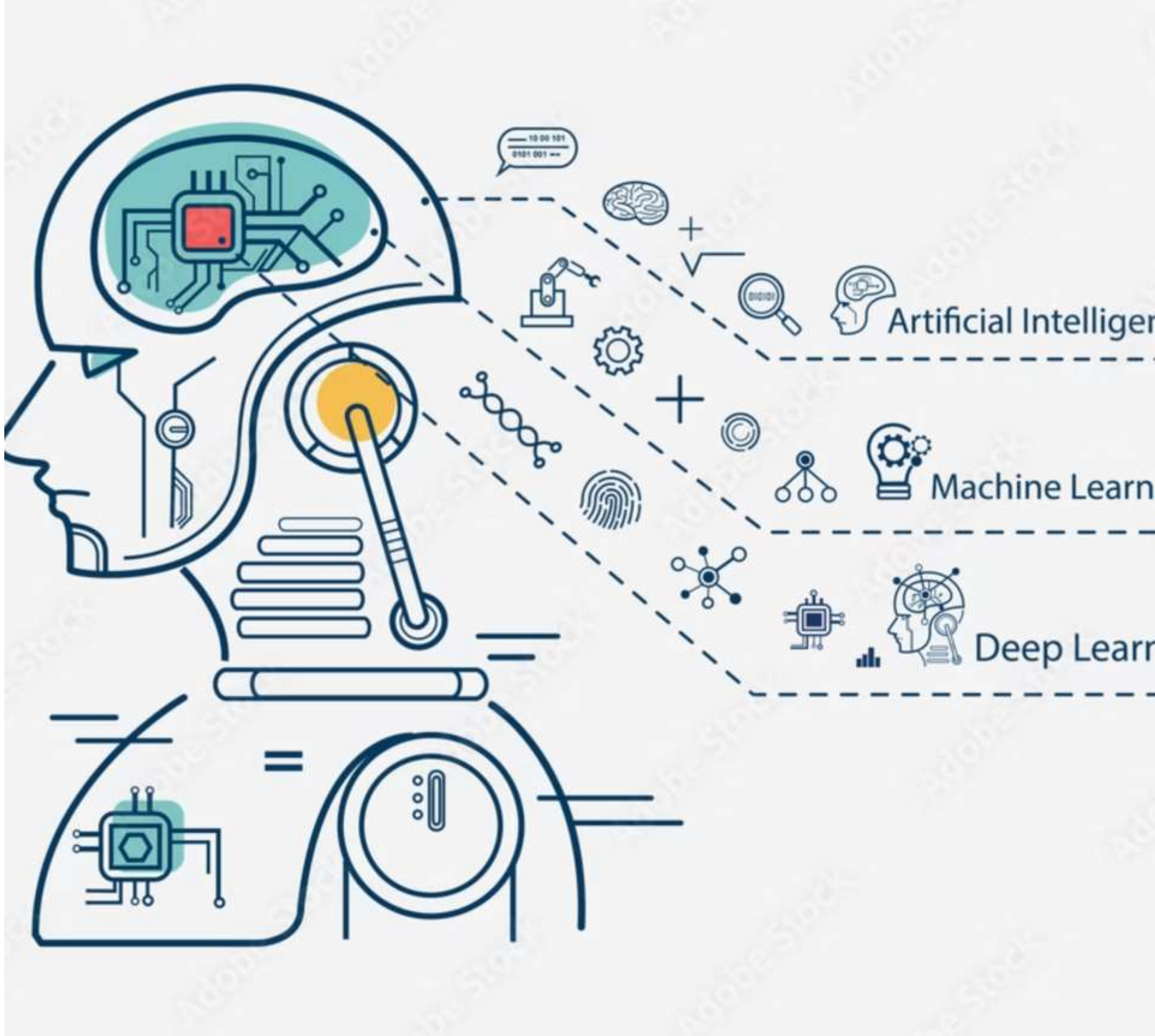

NAVIGATION

# Agenda

01

Cost Function Fundamentals

02

Gradient Descent

03

Local vs. Global Minima

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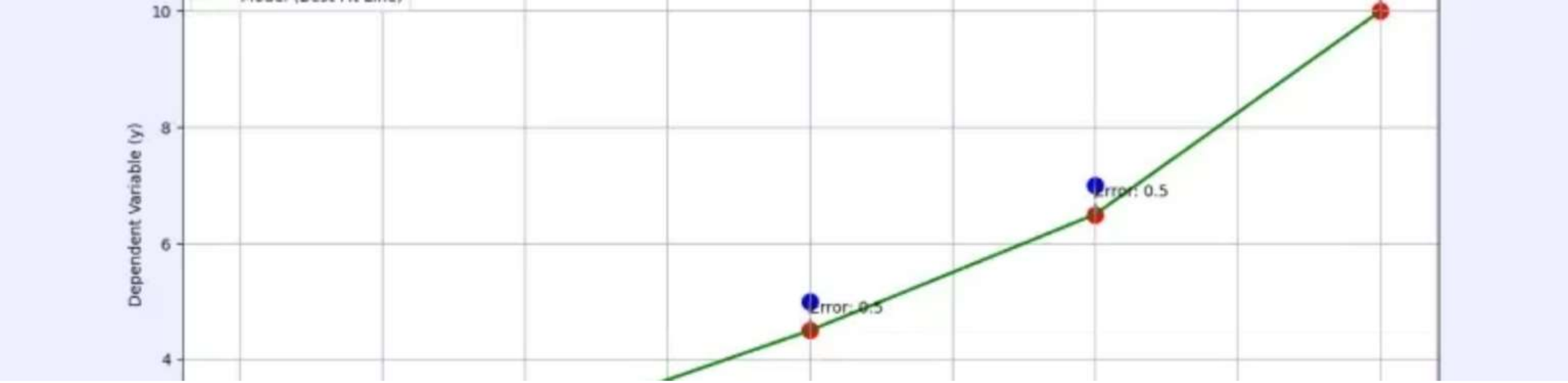


# What is a Cost Function?

A Cost Function is a mathematical function that measures how wrong a model's predictions are.

A cost function measures the difference between the actual value and the predicted value.

If the error is large → the cost will be high

If the error is small → the cost will be low

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# 🎯 Goal in Machine Learning

The main goal in machine learning is **to minimize the cost function or** we also say that we try to improve the model so that its predictions are as close as possible to the actual values.

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# Example: Linear Regression

## Suppose we have a simple linear model:

$y = mx + b$

This is the equation of a straight line.

## What do these mean?

- **y** = predicted output (what the model guesses)
- **m** = slope (how steep the line is)
- **b** = intercept (where the line crosses the y-axis)

The model uses this formula to predict values.

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# Mean Squared Error (MSE)

To check how good our model is, we use **Mean Squared Error (MSE)**.

Formula

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Steps:

1. Take the **actual value** (real answer).

2.Subtract the **predicted value** (model's guess).

3.Square the difference.

4.Add all squared differences.

5.Divide by total number of data points (n).

That final number is the **cost**.

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# Why do we square the error?

We square the error because:

- It removes negative signs (no negative errors).
- Bigger mistakes get more punishment.
- It helps in mathematical calculations.

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# Numerical Example

## Given Data:

| x | Actual y | Predicted y |
|---|---|---|
| 1 | 3 | 2 |
| 2 | 5 | 4 |

# Step 1: Find the error

For first value:

3 − 2 = 1

For second value:

5 − 4 = 1

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# Step 2: Square the errors

$1^2 = 1$ $1^2 = 1$

# Step 3: Take the average

$MSE = (1 + 1) / 2 = 1$ MSE=(1+1)/2=1

# 👉 Final Result:

The cost (MSE) = 1

This means the model has some error.

If the cost becomes closer to 0, the model becomes better.

- Small MSE → Model predictions are close to real values
- Large MSE → Model predictions are far from real values

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# Why is the Cost Function Important?

## 1 It tells the model how wrong it is

The cost function measures the difference between:

- The real answer (actual value)
- The model's answer (predicted value)

If the difference is big → the cost is high.

If the difference is small → the cost is low.

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## 2 Optimization algorithms reduce it

Algorithms like Gradient Descent try to make the cost smaller.

They change the model's settings (parameters) step by step to reduce the error.

## 3 Better cost means better model

If the cost becomes small:

- The model's predictions are close to real answers
- The model works better

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# Why is the Cost Function Important?

## Why Cost Function Important?

1. Model ko batata hai ki woh kitna galat hai
2. Optimization algorithms (like Gradient Descent) isi ko minimize karte hain
3. Better cost → Better model

### ◆ Gradient Descent Relation

Cost function ko minimize karne ke liye hum:

👉 Derivative nikalte hain

👉 Gradient Descent apply karte hain

👉 Parameters update karte hain

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# ◆ Types of Cost Functions

## 1 Regression ke liye:

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

## 2 Classification ke liye:

- Binary Cross Entropy
- Categorical Cross Entropy

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KEY INSIGHT

# Gradient Descent

Gradient Descent is a method used in machine learning to **reduce error**.

When a model makes a prediction, there is some error.

Gradient Descent helps the model **improve step by step** by reducing that error.

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# What is Minimum?

When we use gradient descent, we try to find the **lowest point** of the error graph.

There are two types of minimum:

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# 1 Global Minimum

Global minimum is the **lowest point in the whole graph**.

✓ It is the best possible solution.
✓ Error is the smallest here.

# 2 Local Minimum

Local minimum is a point that looks lowest in a small area,
but it is **not the lowest in the whole graph**.

It is like being stuck in a small pit on a hill,
but there is a deeper valley somewhere else.

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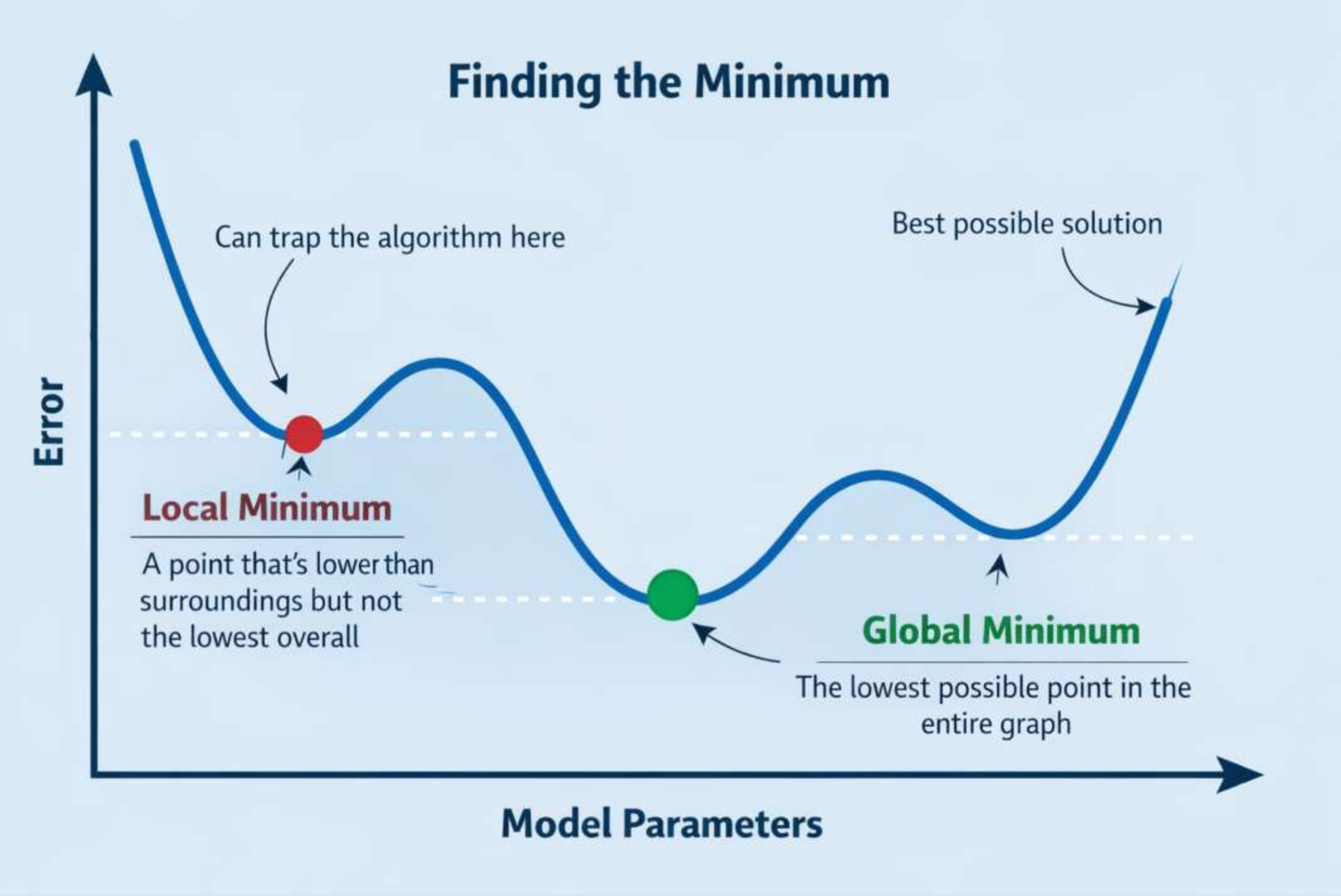
Finding the Minimum
Can trap the algorithm here
Best possible solution
Error
Local Minimum
A point that's lower than surroundings but not the lowest overall
Global Minimum
The lowest possible point in the entire graph
Model Parameters

# 🌟 Why Is This Important?

When we train a machine learning model, our goal is simple:

👉 **Reduce error as much as possible.**👉 Find the **lowest point (minimum)** of the loss function.

But the type of model changes how easy or difficult this is.

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# THANK YOU

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